A determinant criterion for extending generalized splines

MathDB 372959: a reduction to ordinary graph splines

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Task 1: the original problem, solved. Conjecture 1.3 of Dilaver–Altınok [1] asks whether every basis of the extending spline module has determinant associated to its specified element $\widehat{Q}_G$. We prove this for every GCD domain, without the coprimality or PID restrictions in the source. The input is the existing ordinary-spline theorem of Fişekci–Sarıoğlan [2]; the reduction and the exact identification of its determinant factors are given below.

Let G have ordered vertices $v_1, \dots, v_n$. In the source's setting, vertex and edge modules are $m_i R$ and $R/r_e R$, with the natural quotient maps. Thus

$$\widehat{R}_G = \{f \in R^n : m_i \mid f_i \text{ for every } i, \quad r_{ij} \mid f_i - f_j \text{ for every } v_i v_j \in E(G)\}.$$

The labels are nonzero. In fact, the existence of an n-element basis already excludes a zero vertex label or a zero label on an edge between distinct vertices: either imposes a nontrivial linear relation over the fraction field, giving rank less than n. Loops impose no condition and may be omitted. Empty graphs and disconnected graphs are allowed. All gcds, lcms and equalities between their products are understood up to units; the lcm of an empty family is 1.

For distinct vertices put

$$d_{ij} = \operatorname{lcm}_{P : v_i \rightsquigarrow v_j \text{ in } G} \gcd(r_e : e \in P).$$

One may use all trails, as in the source, or just simple paths: removing a closed subwalk increases the path gcd in the divisibility order, and every simple path is a trail. If there is no path, set $d_{ij} = 1$. Definition 3.2 of [1] is precisely

$$q_i = \operatorname{lcm}\big(m_i, \{\gcd(m_j, d_{ij}) : j > i\}, \{d_{is} : s < i\}\big), \qquad \widehat{Q}_G = \prod_{i=1}^{n} q_i. \tag{1}$$

Theorem 1. *For any $F_1, \dots, F_n \in \widehat{R}_G$ over a GCD domain R, these splines form an R-basis if and only if*

$$\det(F_1, \dots, F_n) = u\widehat{Q}_G \qquad (u \in R^\times).$$

In particular, the conjectured necessary condition holds in full.

Add a new vertex v_0, join it to each v_i by an edge labelled m_i, and retain every edge and label of G. Denote the resulting ordinary edge-labelled graph by H, ordered $v_0 < v_1 < \cdots < v_n$. Let $S_H \subseteq R^{n+1}$ be its ordinary spline module.

Lemma 2. *The map*

$$R \oplus \widehat{R}_G \longrightarrow S_H, \qquad (a, f) \longmapsto (a, a + f_1, \dots, a + f_n)$$

is an isomorphism. Moreover, the columns $\mathbf{1}, (0, F_1), \dots, (0, F_n)$ have determinant $\det(F_1, \dots, F_n)$ in the ascending row order.

Proof. For $g \in S_H$, the spoke conditions give $m_i \mid g_i - g_0$, and the old edge conditions are unchanged by subtracting g_0 from each coordinate. The inverse is therefore $g \mapsto (g_0, (g_i - g_0)_{i=1}^n)$. The asserted matrix has first row $(1, 0, \ldots, 0)$; expand along that row. □

Lemma 3. *Let $L_0 = 1$, and for $i \geq 1$ let L_i be the lcm of path gcds in H from v_i to any earlier vertex in $\{v_0, \ldots, v_{i-1}\}$. Then $L_i = q_i$ up to a unit, and the ordinary determinant element $Q_H = \prod_{i=0}^n L_i$ equals $\widehat{Q}_G$ up to a unit.*

Proof. The direct spoke gives $m_i \mid L_i$. Every old path to v_s, $s < i$, is still a path in H, giving $d_{is} \mid L_i$. For $j > i$, append the spoke $v_j v_0$ to each simple path in G from v_i to v_j. These are paths to an earlier vertex. Gcd/lcm distributivity in a GCD domain gives

$$\gcd(m_j, d_{ij}) = \operatorname{lcm}_{P : v_i \rightsquigarrow v_j \text{ in } G} \gcd(m_j, \{r_e : e \in P\}) \mid L_i.$$

The same assertion is vacuous when the path family is empty. Consequently $q_i \mid L_i$.

Conversely, take any simple path in H from v_i to an earlier vertex and truncate it at its first visit to $\{v_0, \ldots, v_{i-1}\}$. The gcd of the original path divides that of this prefix. If the prefix ends at v_s with $1 \leq s < i$, it stays entirely in G, so its gcd divides d_{is}. If it ends at v_0, its last edge is a spoke $v_j v_0$. A direct spoke has gcd m_i. Otherwise $j > i$, and the preceding segment lies in G, so the prefix gcd divides $\gcd(m_j, d_{ij})$. Each case gives a divisor of q_i. Taking the lcm over all paths proves $L_i \mid q_i$. □

Proof of Theorem 1. The ordinary determinant criterion [2, Theorem 4.5, p. 367] states that on any finite edge-labelled graph over a GCD domain, a set of as many splines as vertices is a basis exactly when its determinant is a unit times the product of these earlier-vertex path factors. This is Theorem 5.3 in the arXiv version of that paper.

By Lemma 2, the proposed $F_1, \ldots, F_n$ form a basis of $\widehat{R}_G$ exactly when $\mathbf{1}, (0, F_1), \ldots, (0, F_n)$ form a basis of S_H. Apply the cited theorem on H, use its block determinant, and then use Lemma 3. Reversing the row order, as in the primary source, changes only a sign, which is a unit. □

Scope and verification. No flow-up basis, Bezout identity, prime factorization, or Noetherian assumption is used. In particular, gcd 1 is never mistaken for comaximality. The source's PID and pairwise-coprime cases are existing results, and the ordinary-spline theorem is explicitly credited. The accompanying exact verifier compares independently enumerated path factors, counts finite residue lattices over $\mathbb{Z}$, and checks polynomial-labelled examples. These finite computations are regression checks; the two lemmas prove the assertion for all graphs and GCD domains.

Task 2 (proposed, not solved here). Find an order-independent expression for $\widehat{Q}_G$ in terms of rooted spanning forests of G that remains valid when $\widehat{R}_G$ is not free, and compare its arithmetic complexity with direct path enumeration.

Task 3 (proposed, not solved here). For a cycle G over $k[x, y, z]$, with every vertex and edge label a nonzero linear form, characterize exactly when $\widehat{R}_G$ is free in terms of the linear dependencies among the labels.

Task 4 (proposed, not solved here). Over a Dedekind domain, replace the principal vertex modules $m_i R$ by invertible ideals I_i. Determine the determinant ideal and the Steinitz class of the resulting spline module from the labelled graph.

Task 5 (proposed, not solved here). Replace the edge congruence by $a_{ij}f_i - b_{ij}f_j \in r_{ij}R$, where a_{ij}, b_{ij} are units. Characterize when multiplication of vertex coordinates by units turns this system into an ordinary augmented graph, and obtain the determinant criterion in the remaining cases with nontrivial cycle holonomy.

References

[1] G. Dilaver and S. Altınok, *Basis Criteria for Extending Generalized Splines*, arXiv:2602.04440v1 (2026), Conjecture 1.3 and Definition 3.2.

[2] S. Fişekci and S. Sarıoğlan, *Basis condition for generalized spline modules*, J. Algebraic Combin. **59** (2024), 359–369, doi:10.1007/s10801-023-01290-y; arXiv:2301.12529.